MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, HIGH-SPEED
FLIP-FLOPS, MONOLITHIC SILICON

This limited coordination military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/22B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, high speed, bistable logic microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/022	01	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
02	Dual J-K master-slave flip-flop	54H73
03	Dual D-type edge-triggered flip-flop	54H74

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 Vdc to 7.0 Vdc
Input voltage range	0 to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:55Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and circuit descriptions. The truth tables and descriptions shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I herein.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:55Z
Check the source to verify that this is the currentversion before use.

TABLE I. Electrical characteristics tests (device type 02).

Test No. 1/ 2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2			V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V		0.8		V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OH} = -500 µA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OL} = 20 mA		0.4	±.04	V
5	Low level input current at J, K, or clock	I_{IL1}	V_{CC} = 5.5 V	-1	-2		mA
6	Low level input current at clear	I_{IL2}	V_{IN} = 0.4 V	-2	-4		mA
7*	High level input current at J, K, or clock	I_{IH1}	V_{CC} = 5.5 V		50	±5.0	µA
8*	High level input current at clear	I_{IH2}	V_{IN} = 2.4 V		100	±10.0	µA
9	High level input current at J, K, or clock	I_{IH3}	V_{CC} = 5.5 V		100		µA
10	High level input current at clear	I_{IH4}	V_{IN} = 5.5 V		200		µA
11	Short circuit output current	I_{OS}	V_{CC} = 5.5 V	-40	-100		mA
12	Supply current	I_{CC}	V_{CC} = 5.5 V		50		mA
13	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -8 mA T_A = 25°C		-1.5		V

1/ Delta computations shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics tests (device type 03).

Test No.	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ)	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2			V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V		0.8		V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OH} = -1 mA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OL} = 20 mA		0.4	.04	V
5	Low level input current at preset or D	I_{IL1}	V_{CC} = 5.5 V	-1	-2		mA
6	Low level input current at clear or clock	I_{IL2}	V_{IN} = 0.4 V	-2	-4		mA
7*	High level input current at D	I_{IH1}	V_{CC} = 5.5 V		501	±5	µA
8*	High level input current at preset or clock		V_{IN} = 2.4 V		100	±10	µA
9*	High level input current at clear	I_{IH3}			150	±15	µA
10	High level input current at D	I_{IH4}	V_{CC} = 5.5 V		100		µA
11	High level input current at preset or clock	I_{IH5}	V_{IN} = 5.5 V		200		µA
12	High level input current at clear	I_{IH6}			300		µA
13	Short circuit output current	I_{OS}	V_{CC} = 5.5 V	-40	-100		mA
14	Supply current	I_{CC}	V_{CC} = 5.5 V		50		mA
15	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -8 mA T_A = 25°C		-1.5		V

1/ Delta computations shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:55Z
Check the source to verify that this is the current version before use.

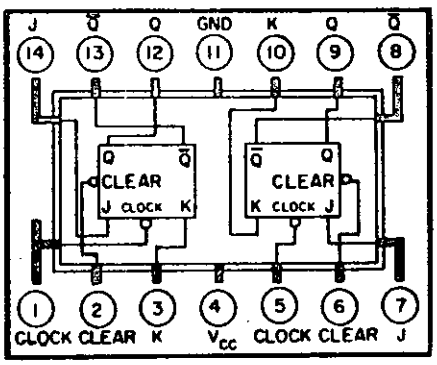
TABLE II. Switching characteristics.

Characteristic 1/	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Device type 02							
Frequency of input count pulses	F_{MAX}	C_L = 50 pF R_L = 280Ω ±5% V_{CC} = 5.0 V	5	25°C	23		MHz
				-55°C 125°C	20		MHz
Propagation delay time high to low level clear to Q	t_{PHL}		6	25°C	2	27	ns
				-55°C 125°C	2	31	ns
Propagation delay time low to high level clear to $\overline{Q}$	t_{PLH}		6	25°C	2	20	ns
				-55°C 125°C	2	24	ns
Propagation delay time high to low level clock to Q or $\overline{Q}$	t_{PHL}		5	25°C	2	34	ns
				-55°C 125°C	2	42	ns
Propagation delay time low to high level clock to Q or $\overline{Q}$	t_{PLH}		5	25°C	2	26	ns
				-55°C 125°C	2	35	ns
Device type 03							
Frequency of input count pulses	f_{MAX}	C_L = 50 pF R_L = 280Ω ±5% V_{CC} = 5.0 V	7,8	25°C	35		MHz
				-55°C 125°C	28		MHz
Propagation delay time high to low level preset to $\overline{Q}$	t_{PHL}		6	25°C	2	32	ns
				-55°C 125°C	2	38	ns
Propagation delay time low to high level preset to Q	t_{PLH}		6	25°C	2	20	ns
				-55°C 125°C	2	27	ns
Propagation delay time high to low level clear to Q	t_{PHL}		6	25°C	2	32	ns
				-55°C 125°C	2	38	ns
Propagation delay time low to high level clear to $\overline{Q}$	t_{PLH}		6	25°C	2	20	ns
				-55°C 125°C	2	27	ns
Propagation delay time high to low level clock to Q or $\overline{Q}$	t_{PHL}		7,8	25°C	2	22	ns
				-55°C 125°C	2	25	ns
Propagation delay time low to high level clock to Q or $\overline{Q}$	t_{PLH}		7,8	25°C	2	17	ns
				-55°C 125°C	2	22	ns

1/ Test each flip-flop.

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 02



Device type 03

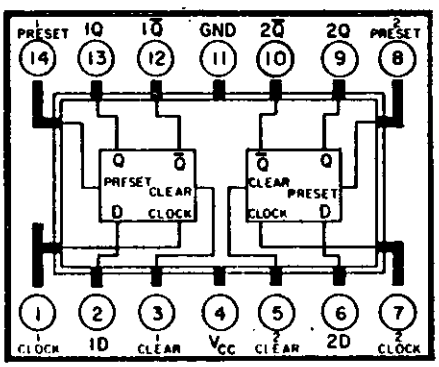


FIGURE 1. Logic diagrams and terminal connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:55Z
Check the source to verify that this is the current version before use.

Device 02

Truth table each flip-flop		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to clear sets Q to low-level.
Clear is independent of clock.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

Device 03

Truth table each flip-flop		
t_n	t_{n+1}	
INPUT D	OUTPUT Q	OUTPUT $\overline{Q}$
L	L	H
H	H	L

Positive logic: Low input to preset sets Q to high-level.
Low input to clear sets Q to low-level.
Clear and preset are independent of clock.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

FIGURE 2. Truth tables and logic equations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 02

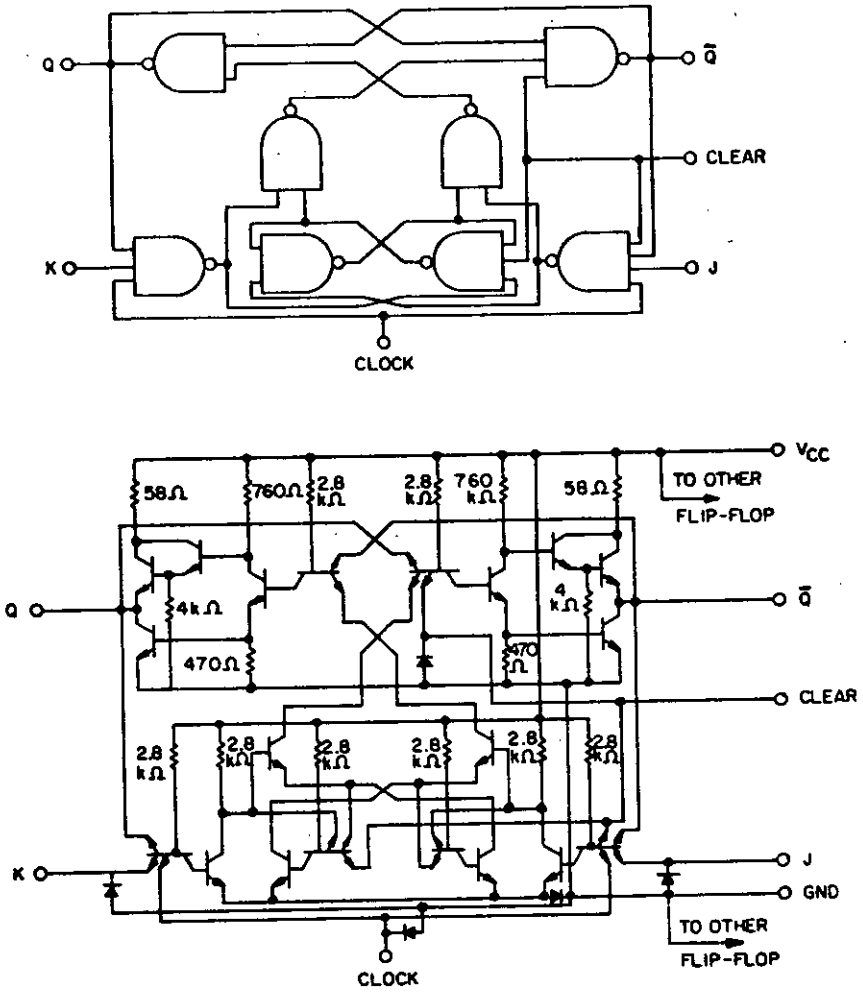
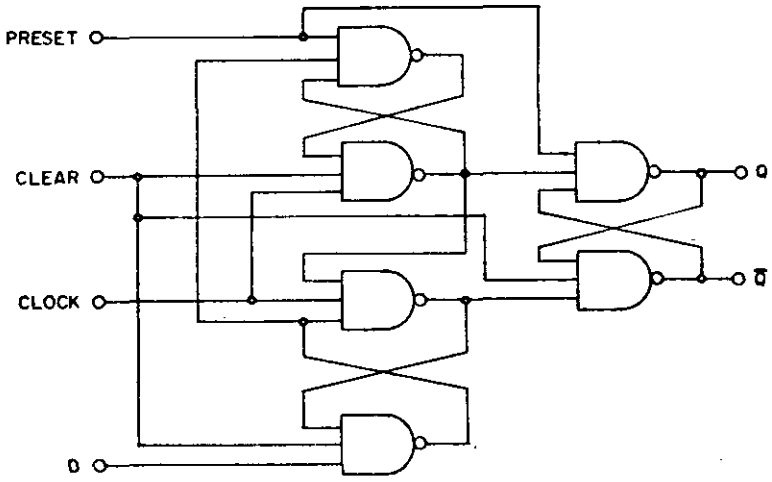


FIGURE 3. Schematic circuits and logic diagrams.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 03



FUNCTIONAL BLOCK DIAGRAM

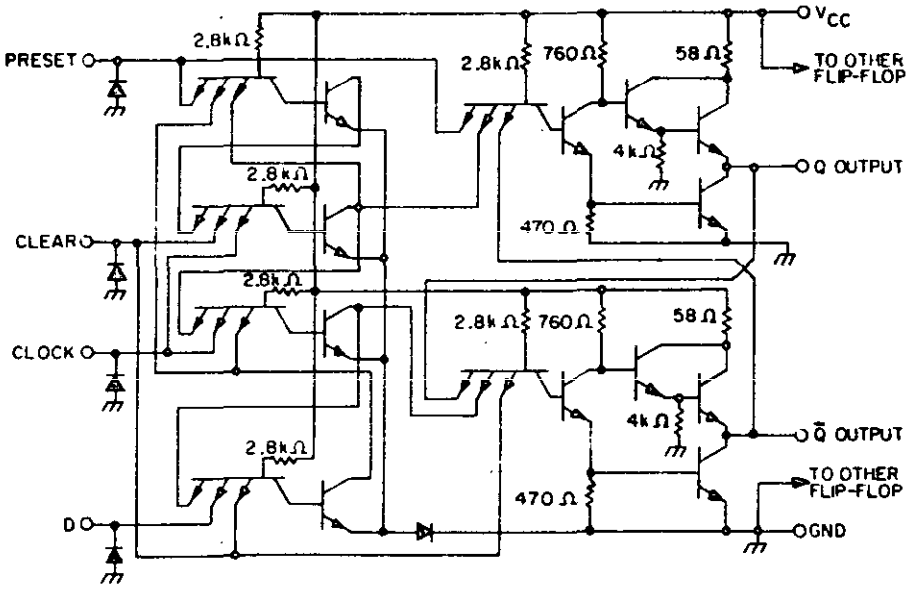
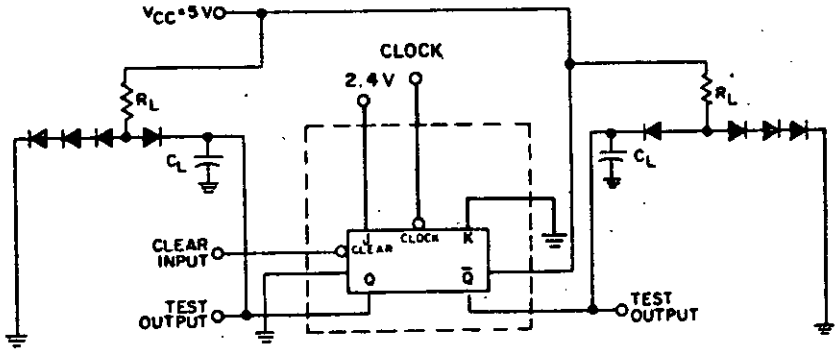


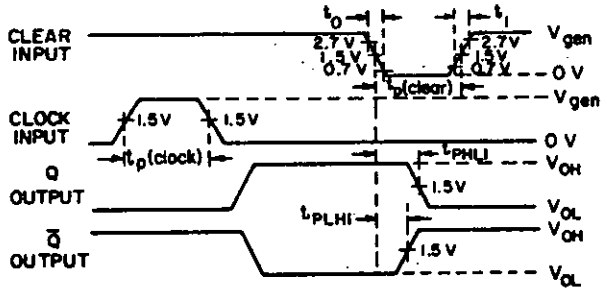
FIGURE 3. Schematic circuits and logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 02



TEST CIRCUIT



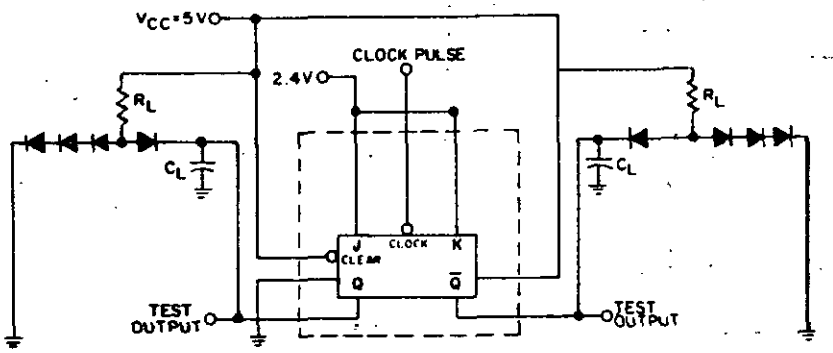
VOLTAGE WAVEFORMS

NOTES:
1. Clear inputs dominate regardless of the state of clock or J-K inputs.
2. Clear input pulse characteristics: V_{gen} = 3 V, $t_0 = t_1$ = 7 ns, $t_{p(clear)}$ = 16 ns, PRR = 1 MHz.
3. All diodes are 1N3064 or equivalent.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 280Ω ±5%.
6. Clock input pulse characteristics: V_{gen} = 3 V, $t_{p(clock)} \geq$ 20 ns, PRR = 1 MHz.

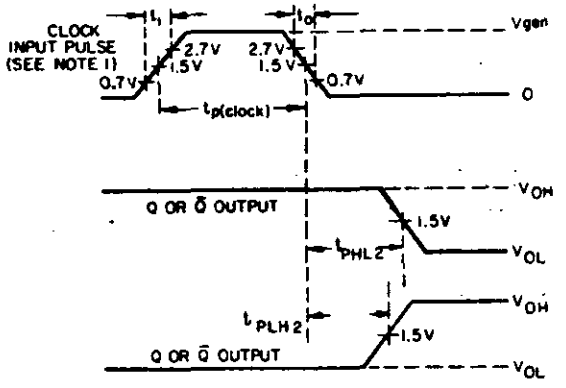
FIGURE 4. Clear switching test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 02



TEST CIRCUIT



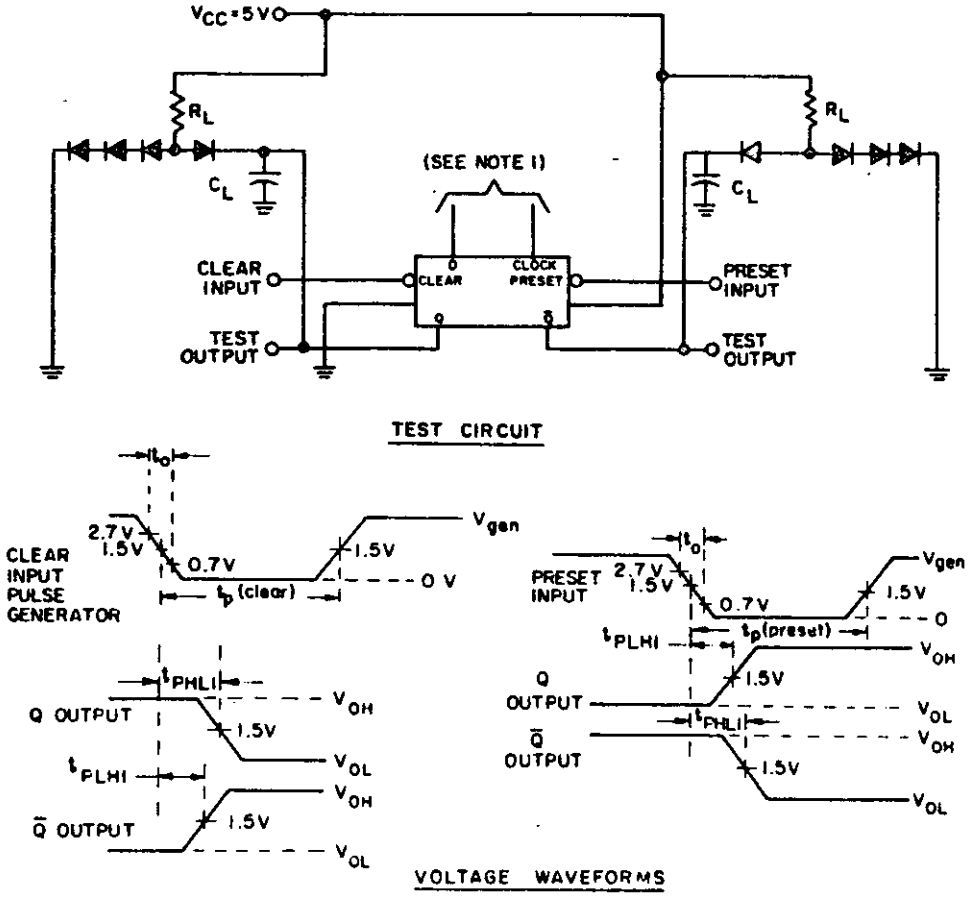
VOLTAGE WAVEFORMS

NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), V_{gen} = 3 V, t_1 = t_0 = 7 ns, t_p (clock) = 20 ns, and PRR = 1 MHz. All J and K inputs are at 2.4 V. When testing f_{MAX} the clock input characteristics are V_{gen} = 3 V, t_1 = t_0 = 3 ns, t_p (clock) = 12 ns, and PRR = 1 MHz.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF minimum (including jig and probe capacitance).
4. R_L = 280Ω ±5%.

FIGURE 5. Synchronous switching test circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 03

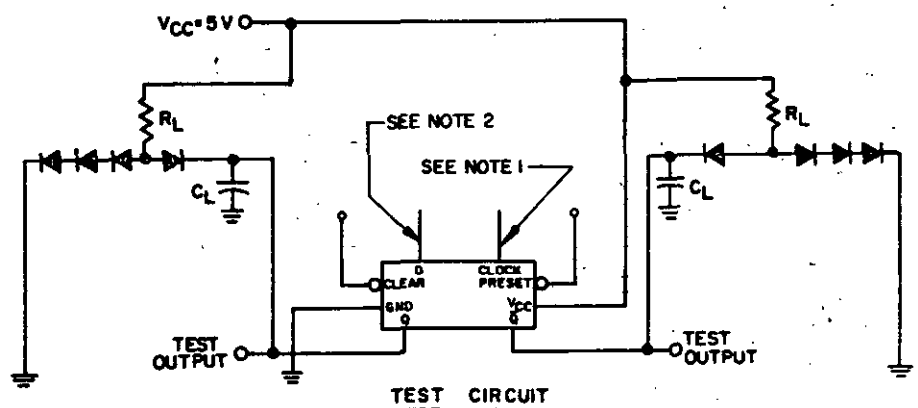


NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or D inputs.
2. All diodes are 1N3064, or equivalent.
3. Clear or preset input pulse characteristics: V_{gen} = 3 V, $t_0 \leq 7$ ns, t_p(clear) = t_p(preset) = 25 ns, and PRR = 1 MHz.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 280Ω ±5%.

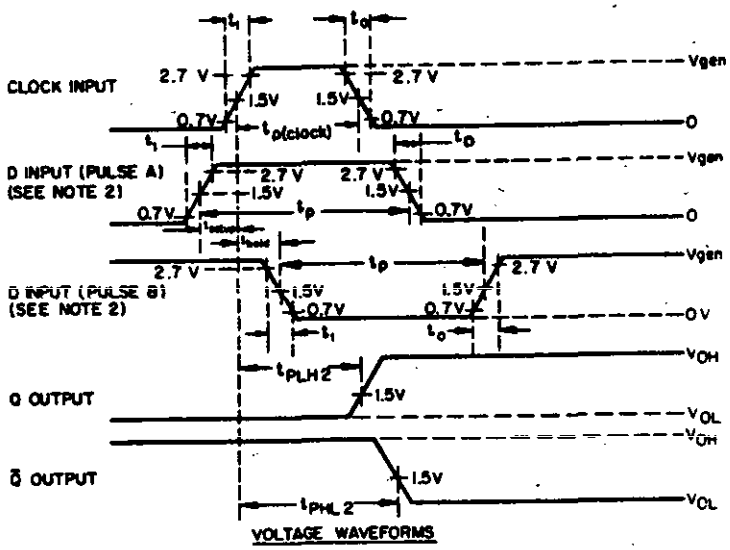
FIGURE 6. Clear and preset switching test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 03



TEST CIRCUIT



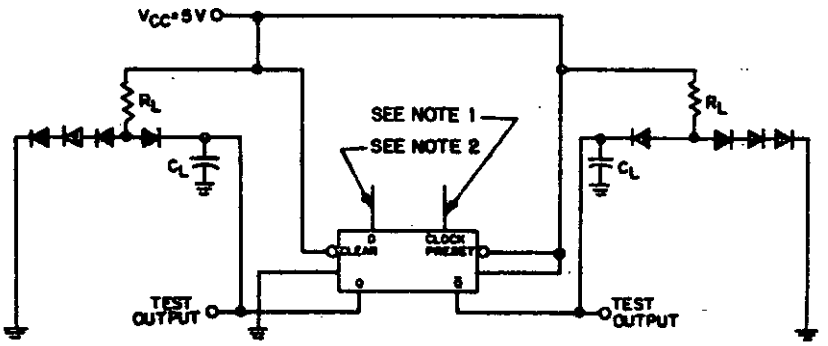
VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 7$ ns, t_p(clock) = 20 ns, and PRR = 1 MHz. When testing f_{MAX}, PRR = see table III.
2. D input (pulse A) has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 7$ ns, t_{setup} = 10 ns, t_p = 60 ns, and PRR is 50% of the clock PRR. D input (pulse B) has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 7$ ns, t_{hold} = 5 ns, t_p = 60 ns, and PRR is 50% of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 280Ω ±5%.

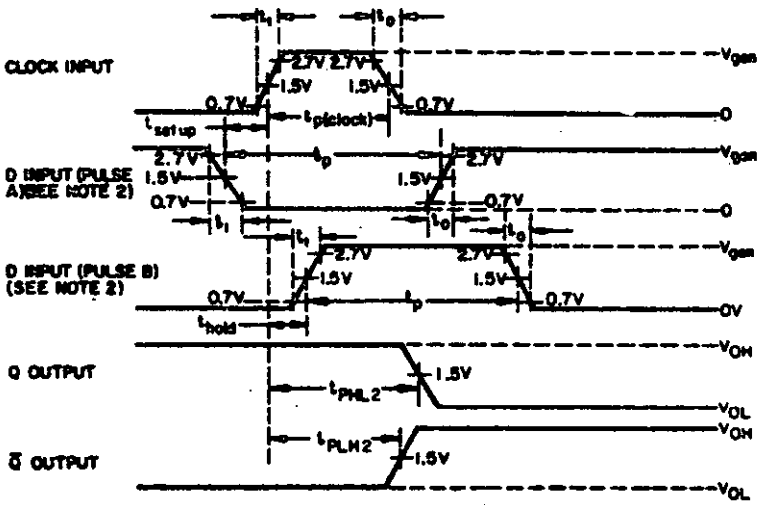
FIGURE 7. Synchronous switching test circuit (high-level data).

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the current version before use.

Device type 03



TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 7$ ns, t_p(clock) = 20 ns, and PRR = 1 MHz. When testing f_{MAX}, PRR = see table III.
2. D input (pulse A) has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 7$ ns, t_{setup} = 15 ns, t_p = 60 ns, and PRR is 50% of the clock PRR. D input (pulse B) has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 7$ ns, t_{hold} = 5 ns, t_p = 60 ns, and PRR is 50% of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 280Ω ±5%.

FIGURE 8. Synchronous switching test circuit (low-level data).

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:55Z
Check the source to verify that this is the currentversion before use.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figures 4 and 5 shall be performed at 25 C. All devices shall meet the limits specified in table II.

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-57)

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:55Z
Check the source to verify that this is the current version before use.